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DATA ANALYTICS

Database : employees_db

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San francisco

① SELECT Statement

Write a SQL query to retrieve all columns from the employees table

```
SELECT *
FROM `employees_db`.`default`.`employees`;
```

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San francisco

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② SELECT DISTINCT Statement

Write a SQL query to find all the unique departmen in the employee table

```
SELECT DISTINCT (department)
FROM `employee_db`.`default`.`employees`;
```

	department
1	IT
2	HR
3	Finance
4	Marketing

③ ORDER BY Statement

Write a SQL query to retrere all employees first and last names, ordered by salary in descending order

```
SELECT first_name, last_name, Salary
FROM `employee_db`.`default`.`employees`
ORDER BY Salary DESC;
```

	first_name	last_name	salary
1	Emily	Davis	62000
2	Mike	Johnson	60000
3	Robert	Wilson	59000
4	John	Doe	55000
5	Sarah	Brown	53000
6	Daniel	Clark	53000
7	David	White	52000
8	Jessica	Moore	51000
9	Laura	Hall	50000
10	Jane	Smith	48000

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④ LIMIT Statement

Write a SQL query to retrieve the top 5 highest paid employees

```
SELECT first_name, last_name, department, Salary
FROM `employee_db`.`default`.`employees`
ORDER BY Salary DESC
LIMIT 5;
```

	first_name	last_name	department	Salary
1	Emily	Davis	IT	62000
2	mike	Johnson	Finance	60000
3	Robert	wilson	Finance	59000
4	John	Doe	IT	55000
5	Sarah	Brown	TI	53000

⑤ WHERE Statement

Write a SQL query to find employees who work in the IT department

```
SELECT first_name, last_name, department
FROM `employee_db`.`default`.`employees`
WHERE department = 'IT';
```

first_name	last_name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

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⑥ AND Statement

Write a SQL query to find employees who work in the finance department AND have salary greater than 58000

```
SELECT first_name, last_name, department, Salary
FROM 'employee_db', 'default', 'employees'
WHERE department = 'finance'
AND Salary > 58000;
```

	first_name	last_name	department	Salary
1	mike	Johnson	Finance	60000
2	Robert	Wilson	Finance	59000

⑦ OR Statement

Write a SQL query to find employee who work in the HR department OR the marketing department

	first_name	last_name	department
1	Jane	Smith	HR
2	David	White	Marketing
3	Jessica	Moore	HR
4	Daniel	Clark	Marketing

```
SELECT First_name, last_name, department
FROM `employee_db`, `default`, `employees`
WHERE department = `HR` ~~OR~~
OR department = `marketing`;
```

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⑧ NOT Statement

Write a SQL query to find employees who do not work in the IT department

```
SELECT first_name, last_name, department
FROM `employee_db`.'default'.'employees'
WHERE NOT department = `IT`;
```

	first_name	last_name	department
1	Jane	Smith	HR
2	Mike	Johnson	Finance
3	David	White	Marketing
4	Robert	Wilson	Finance
5	Jessica	Moore	HR
6	Daniel	Clark	Marketing

⑨ IN Statement

Write a SQL statement to find employees who are in the HR, IT or Finance departments

```
SELECT first_name, last_name, department
FROM `employee_db`.'default'.'employees'
WHERE department IN ('HR', 'IT', 'finance');
```

	first_name	last_name	department
1	John	~~Johnson~~ Doe	~~Finance~~ IT
2	Jane	~~Robert~~ Smith	HR
3	Mike	~~Davis~~ Johnson	Finance
4	Sarah	Brown	IT
5	Emily	Davis	IT
6	Robert	Wilson	Finance
7	Jessica	Moore	HR
8	Laura	Hall	IT

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(10) Combining Conditions

Write a SQL query to find employees who are in the IT department, have a salary greater than 50 000 and located in New York

```
SELECT id, first_name, last_name, department, city
FROM `~~work~~ employee_db` 'default'.'employees'
WHERE department = 'IT' AND salary > 50000
AND city = 'New York';
```

	id	first_name	last_name	department	city
1	1	John	Doe	IT	New York
2	4	Sarah	Brown	IT	New York

(11) Combining WHERE, AND, and ORDER BY

Write a SQL query to retrieve the first and last_name of empl who works in the Finance or Marketing department, earn more than 52 000 and order the results by salary in descending o

```
SELECT first_name, last_name, department, salary
FROM `~~work~~ _employee_db` 'default'.'employees'
WHERE department IN ('finance', 'marketing')
AND salary > 52000
ORDER BY salary DESC;
```

	first_name	last_name	department	salary
1	Mike	Johnson	finance	60000
2	Robert	Wilson	finance	59000
3	Daniel	Clark	marketing	53000

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(12) Combining SELECT DISTINCT, WHERE, and IN

Write a SQL to find all the unique cities where employee work, excluding those in the IT and HR department

```
SELECT DISTINCT city, Department
FROM 'employee_db'.'default'.'employees'
WHERE department NOT IN ('IT', 'HR');
```

	city	Department
1	Los Angeles	Finance
2	San Francisco	Marketing
3	Houston	Finance
4	Chicago	Marketing

(13) Combining WHERE, NOT, AND, and ORDER BY

Write a SQL to retrieve employees who are NOT in the finance department, have a salary greater than 50000 and order the results by hire date in ascending order

```
SELECT first_name, last_name, salary
FROM 'employee_db'.'default'.'employees'
WHERE NOT department = 'finance'
AND salary > 50000
ORDER BY hire_date ASC;
```

first_name	last_name	Salary
Emily	Davis	62000
David	White	52000
Jessica	Moore	51000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000

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(14) Combining WHERE, OR, IN, and LIMIT

Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department

```
SELECT first_name, last_name, city, department
FROM 'employee_db'.'default'.'employees
WHERE city = 'chicago'
OR city = 'Los Angeles'
AND department IN ('IT', 'Marketing')
LIMIT 3;
```

	first_name	last_name	city	department
1	Jane	Smith	Chicago	HR
2	Emily	Davis	Chicago	IT
3	Daniel	Clark	Chicago	Marketing

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